

Kulin Mathur

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Profile

Aspiring Computer Science graduate specializing in Artificial Intelligence and Machine Learning with hands-on experience building production-ready NLP systems, predictive models, and full-stack web applications. Proficient in Python, machine learning pipelines, retrieval-augmented generation (RAG), and REST API development. Passionate about emerging technologies including blockchain and large language models (LLMs).

Education

Vellore Institute of Technology 2023 – Present
B.Tech in Computer Science and Engineering (AI/ML Specialization), GPA: 8.89/10 Chennai, India

RBSE 2021 – 2022
12th Grade, Percentage: 85.4% Rajasthan, India

CBSE 2019 – 2020
10th Grade, Percentage: 88% Rajasthan, India

Technical Skills

Languages: Python, Java, C, C++, SQL, NoSQL, JavaScript, HTML, CSS
AI / ML: Machine Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Sentence-BERT, FAISS, scikit-learn, Statistics
Core CS: Data Structures and Algorithms, Object-Oriented Programming, REST API Design
Frameworks / Tools: FastAPI, Flask, React.js, Node.js, Express.js, MongoDB, Docker, Git, Linux

Experience

Web Development Intern 2025
InAmigos Foundation Remote

- Completed a focused 2-week web development internship, designing and deploying a multi-page organizational website (Home, About, Programs, Get Involved, Contact) using Wix AI based on real organizational data and branding requirements.
- Designed a complete UI/UX system in Figma for the organization's website, including donation workflows, volunteer intake forms, impact showcase sections, and a location-based assistance finder, delivered as publish-ready desktop and mobile prototypes.
- Collaborated with team members across planning, design, and implementation phases to translate organizational requirements into functional digital solutions.
- Strengthened practical skills in website development, UI/UX design, requirement gathering, and professional project workflows through real-world deliverables and stakeholder communication.

Projects

YouTube RAG Question-Answering System 2025
Python, FastAPI, Sentence-BERT, FAISS, Gemini API

- Designed and deployed a production-ready Retrieval-Augmented Generation (RAG) backend that answers natural language questions from YouTube video transcripts with sub-15-second end-to-end latency.
- Implemented a hybrid retrieval pipeline combining FAISS dense vector search and BM25 sparse retrieval, fused via Reciprocal Rank Fusion (RRF), followed by a cross-encoder reranker (ms-marco-MiniLM-L-6-v2) and Maximal Marginal Relevance (MMR) diversity selection; achieved Context Precision@K of 1.00 and MRR of 1.00 across evaluation samples.

- Engineered a 4-method transcript extraction fallback chain (youtube-transcript-api, yt-dlp, Whisper ASR) enabling support for any YouTube video including uncaptioned content; built 7-class semantic intent detection using Sentence-BERT with 93%+ confidence on test queries.
- Exposed REST API via FastAPI with interactive Swagger UI (/docs); evaluated system across 14 metrics including Faithfulness (0.75), Answer Relevancy (0.79), Hallucination Rate (0.25), and Semantic Similarity (0.84).

Loan Approval Prediction System

2024

Python, scikit-learn, Flask, Docker

- Built and deployed a machine learning web application that predicts loan approval using applicant features including credit history, income, loan amount, and property area.
- Trained classification model using scikit-learn; containerized the full application with Docker and deployed a live demo accessible via Flask backend.
- Designed the end-to-end UI/UX for data input and result display; evaluated model performance using accuracy, precision, recall, and confusion matrix analysis.

Involvement

Member

2024 – 2025

Game Development Club, VIT Chennai

Chennai, India

- Applied new game development tools and frameworks through hands-on projects and technical workshops.
- Collaborated in cross-functional teams to design and ship group game projects, strengthening communication and agile teamwork skills.

Certifications

IBM Blockchain Foundations – distributed ledger technology, smart contracts, and enterprise blockchain applications.

Cisco Packet Tracer – network simulation, device configuration, and troubleshooting.

The Complete Full-Stack Web Development Bootcamp – front-end (HTML, CSS, JavaScript, React.js) and back-end (Node.js, MongoDB).

Languages

English (Fluent), Hindi (Fluent)